

Common Arithmetic Fibers of Stably Inequivalent Keller Maps

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Abstract

We show that a complete regular arithmetic fiber does not determine its ambient polynomial Keller map, even up to polynomial left-right equivalence and identity stabilization. For every $N \geq 4$, two fixed Keller maps over $\mathbb{Q}$, of geometric degree N , share the complete fiber

$$\operatorname{Spec} \mathbb{Q}[T]/(T^N + T^3 - 2T^2 + T + u)$$

for every squarefree rational specialization u , but their stable classes are distinct. A Hilbert subset of specializations gives connected fibers. Over $K = \mathbb{Q}(\sqrt{-2})$, three fixed pairwise stably inequivalent quartic Keller maps share every squarefree member of one quartic pencil; again a Hilbert subset is connected. The separation is intrinsic: the normalized ramified boundary strata have different unit ranks, and the two reciprocal constructions are distinguished by a relative Fitting divisor, or independently by boundary-contact thickness. We give small irreducible quartic examples over both fields and exact finite-field certificates.

1 The question and the results

A polynomial map $F : \mathbb{A}_K^n \rightarrow \mathbb{A}_K^n$ over a characteristic-zero field is a *Keller map* if its Jacobian determinant is a nonzero constant. We call a fiber complete if it is regular and its length equals the geometric degree of F . Thus no generic inverse sheet is lost at the boundary.

Two maps are polynomially left-right equivalent if

$$F_2 = \beta \circ F_1 \circ \alpha$$

for polynomial automorphisms α and β of source and target. They are *stably* equivalent if their products with identity maps on affine spaces become left-right equivalent.

For a finite étale K -algebra A , define

$$\mathcal{R}_K(A) = \left\{ \begin{array}{l} \text{stable polynomial left-right classes of Keller maps} \\ \text{having a complete fiber isomorphic to } \operatorname{Spec} A \end{array} \right\}. \quad (1.1)$$

The definition is unmarked: the target carrying the fiber may vary. Completeness makes the condition invariant under left-right equivalence, and stabilization replaces the fiber by its product with the origin.

Our first result holds in every degree.

Theorem 1.1 (Fixed pairs over $\mathbb{Q}$). *For every $N \geq 4$, there are two fixed $\mathbb{Q}$ -defined Keller maps*

$$F_N^{\text{wt}}, F_N^{\text{quad}} : \mathbb{A}^3 \longrightarrow \mathbb{A}^3$$

of geometric degree N with the following properties.

1. For every $u \in \mathbb{Q}$ such that

$$P_{N,u}(T) = T^N + T^3 - 2T^2 + T + u$$

is squarefree, both maps have a complete fiber isomorphic to $\text{Spec } \mathbb{Q}[T]/(P_{N,u})$.

2. The two maps are not polynomially left-right equivalent, even after arbitrary identity stabilization.

3. The polynomial $P_{N,U}$ is irreducible over $\mathbb{Q}(U)$. Consequently a Hilbert subset of $u \in \mathbb{Q}$ gives connected common fibers, and for each such u ,

$$|\mathcal{R}_{\mathbb{Q}}(\mathbb{Q}[T]/(P_{N,u}))| \geq 2.$$

The second result replaces a pair by a triple.

Theorem 1.2 (Fixed triples over $\mathbb{Q}(\sqrt{-2})$). *Let $K = \mathbb{Q}(\eta)$, where $\eta^2 = -2$, and put*

$$\Psi_R(T) = T - \frac{9}{2}T^2 + (8 + 2\eta)T^3 - \left(\frac{7}{2} + 2\eta\right)T^4 - R. \quad (1.2)$$

There are three fixed K -defined Keller maps

$$F^{\text{wt}}, F^{\text{can}}, F^{\text{quad}} : \mathbb{A}_K^3 \longrightarrow \mathbb{A}_K^3$$

of geometric degree four such that:

1. *for every $R \in K$ for which Ψ_R is squarefree, all three maps have a complete fiber isomorphic to $\text{Spec } K[T]/(\Psi_R)$;*
2. *the three maps are pairwise stably polynomially left-right inequivalent;*
3. *Ψ_R is irreducible over $K(R)$, so a Hilbert subset of $R \in K$ gives connected common fibers and*

$$|\mathcal{R}_K(K[T]/(\Psi_R))| \geq 3.$$

The maps are explicit, but the common fibers are best understood through their one-variable inverse equations. We recall the two programmable constructions next, apply them to the pencils (1) and (1.2), and finish with the stable boundary invariants.

2 Two programmable inverse constructions

2.1 The weighted tangent construction

Let $P \in K[T]$ be squarefree of degree at least three, and set

$$H(T) = P(T) - P(0) - P'(0)T.$$

The polynomial P is *tangent-admissible* if

$$P(1) - P(0) = P'(0), \quad c := P'(0) - P'(1) \neq 0, \quad P''(1) \neq 2(P'(1) - P'(0)). \quad (2.1)$$

Only H and c , not the constant term of P , enter the map.

Write $p = H'$, $\kappa = H''(1)/c$, and

$$a = -\frac{1+\kappa}{2+\kappa}, \quad L_H(W) = \int_0^W \frac{VH''(V)}{c} dV.$$

In source coordinates put

$$u = 1 + xy, \quad \gamma = 1 + axy + x^2z, \quad W = u\gamma.$$

The weighted map is

$$F_H^{\text{wt}}(x, y, z) = \left(\frac{u + L_H(W)/\gamma^2}{x^2}, \frac{c + p(W)/\gamma}{x}, x\gamma \right). \quad (2.2)$$

The endpoint identities in (2.1) make the displayed quotients polynomial.

Proposition 2.1 (Weighted inverse). *The map (2.2) satisfies*

$$\det DF_H^{\text{wt}} = c.$$

Writing its target coordinates as (A, B, C) , its inverse equation is

$$E_{A,B,C}(W) = H(W) - BCW + cAC^2. \quad (2.3)$$

At

$$(A, B, C) = \left(\frac{P(0)}{c}, -P'(0), 1 \right),$$

equation (2.3) is exactly $P(W) = 0$. If P is squarefree, reconstruction identifies the complete scheme fiber with $\text{Spec } K[W]/(P)$.

Proof. The weighted tangent identities are

$$BC = H'(W) + c\gamma, \quad cAC^2 = W(H'(W) + c\gamma) - H(W).$$

Eliminating γ gives (2.3). Its derivative gives

$$\gamma = -E'_{A,B,C}(W)/c, \quad x = -cC/E'_{A,B,C}(W).$$

Thus every simple root reconstructs exactly one finite source point. The same two identities reduce the three-variable Jacobian to the plane tangent determinant and give $\det DF_H^{\text{wt}} = c$. $\square$

For the stable comparison we need one additional open condition: H is *boundary-clean* when zero is an exact double root and every nonzero root is simple.

2.2 The root-engineered quadratic gauge

Let

$$G(S) = g_1S + \cdots + g_NS^N, \quad g_1g_3g_N \neq 0.$$

Put

$$t = 1 + xy, \quad q = t^2z + \frac{g_1}{g_3}y^2(1 + 3t).$$

The root-engineered quadratic-gauge map is

$$\begin{aligned}\Pi &= tq, \\ B &= y + 3\frac{g_3}{g_1}xq + 2\frac{g_2}{g_1}tq + \sum_{k=4}^N k\frac{g_k}{g_1}t^2x^{k-2}q^k, \\ C &= x(5 - 3t) - \frac{g_3}{g_1}x^3z - \sum_{k=4}^N (k-2)\frac{g_k}{g_1}(xq)^k.\end{aligned}\tag{2.4}$$

Proposition 2.2 (Quadratic-gauge inverse). *The map (2.4) satisfies $\det DF_G^{\text{quad}} = -2$. Define*

$$G_\Pi(S) = g_1S + \Pi(g_2S^2 + g_3S^3) + \sum_{k=4}^N g_k\Pi^k S^k.$$

Its inverse equation is

$$G_\Pi(S) - \frac{g_1}{2}(BS^2 + C) = 0.\tag{2.5}$$

At $(\Pi, B, C) = (1, 0, 0)$ this is $G(S) = 0$. More generally, changing only C adds an arbitrary constant to the inverse polynomial. Every simple root reconstructs exactly one source point.

Proof. On $t \neq 0$, use $S = x/t$ and $Q = y + xq$. Then

$$D = 1 - SQ + \Pi S^2 = t^{-1}.$$

Differentiating (2.5) on the source incidence gives

$$\partial_S E = g_1 D.$$

The rational source chart contributes the reciprocal factor D^{-1} ; the plane incidence contributes $-2D$. Their product gives determinant -2 , and $D \neq 0$ gives unique reconstruction for every simple root. $\square$

3 Two fixed maps with infinitely many common fibers

Fix $N \geq 4$, and set

$$H_N(T) = T^N + T^3 - 2T^2, \quad P_{N,u}(T) = H_N(T) + T + u.\tag{3.1}$$

The basic arithmetic identities are independent of u :

$$\begin{aligned}P_{N,u}(1) - P_{N,u}(0) &= P'_{N,u}(0) = 1, \\ P'_{N,u}(0) - P'_{N,u}(1) &= 1 - N \neq 0, \\ P''_{N,u}(1) &= N(N-1) + 2 \neq 2(N-1).\end{aligned}$$

Thus every squarefree member is tangent-admissible, and all members use the same weighted map $F_{H_N}^{\text{wt}}$. Proposition 2.1 recovers $P_{N,u}$ at

$$q_u^{\text{wt}} = \left(\frac{u}{1-N}, -1, 1 \right).\tag{3.2}$$

The fixed polynomial

$$G_N(T) = H_N(T) + T = T^N + T^3 - 2T^2 + T$$

satisfies the hypotheses of Proposition 2.2. Its quadratic-gauge map recovers $P_{N,u}$ at

$$q_u^{\text{quad}} = (1, 0, -2u). \quad (3.3)$$

Lemma 3.1 (Boundary cleanliness). *The seed H_N is boundary-clean for every $N \geq 4$.*

Proof. One has

$$H_N(T) = T^2(T^{N-2} + T - 2).$$

The last factor is nonzero at zero and vanishes at one. If it shared a root with its derivative, then

$$T^{N-3} = -\frac{1}{N-2}$$

and substitution into $T^{N-2} + T - 2 = 0$ would give

$$T = \frac{2(N-2)}{N-3} > 0,$$

a contradiction. □

Lemma 3.2 (Generic connectedness). *The polynomial $P_{N,U}$ is irreducible over $\mathbb{Q}(U)$.*

Proof. It is primitive in $\mathbb{Q}[T, U]$ and has degree one in U . In any factorization, one factor has U -degree zero. Comparing the coefficient of U makes that factor a unit in $\mathbb{Q}[T]$. Gauss's lemma finishes the proof. □

Hilbert irreducibility applied to Lemma 3.2 gives a Hilbert subset of rational u with $P_{N,u}$ irreducible [4, Chapter 3]. Propositions 2.1 and 2.2 give the two complete fibers, while Lemma 3.1 permits the stable unit-rank separation in Section 5. This proves Theorem 1.1.

Proposition 3.3 (A small rational quartic). *Let*

$$P(T) = 2T^4 - T^3 - T^2 + T + 1, \quad A = \mathbb{Q}[T]/(P).$$

Then A is a quartic field and $|\mathcal{R}_{\mathbb{Q}}(A)| \geq 2$.

Proof. The tangent data are

$$\begin{aligned} P(1) - P(0) &= P'(0) = 1, & P'(0) - P'(1) &= -3, \\ P''(1) &= 16 \neq 6 = 2(P'(1) - P'(0)). \end{aligned}$$

The weighted seed

$$P(T) - 1 - T = T^2(T - 1)(2T + 1)$$

is boundary-clean, and the weighted target is $(-1/3, -1, 1)$. The quadratic-gauge seed $P - 1$ has $(g_1, g_3, g_4) = (1, -1, 2)$, and target $(1, 0, -2)$.

Modulo 3, multiplication by the inverse leading coefficient gives

$$\overline{P} = T^4 + T^3 + T^2 - T - 1.$$

It has no root in $\mathbb{F}_3$. The only monic irreducible quadratic over $\mathbb{F}_3$ is $T^2 + 1$, and division leaves remainder $T - 1$. Thus P is irreducible. Its discriminant is 1556, so the common fiber is connected and etale. □

4 Three fixed maps with infinitely many common fibers

Let $K = \mathbb{Q}(\eta)$, $\eta^2 = -2$, and consider Ψ_R from (1.2). It is already the inverse polynomial of a fixed quartic cancellation map.

For the type-(2, 1) cancellation construction, put

$$A_0 = 1 + xy^2, \quad B_0 = A_0^2 z + y^3 h(A_0), \quad \Pi = A_0 B_0, \quad Q = y + x B_0,$$

and

$$\mathcal{R} = \int_0^{x/A_0} (1 - t(Q - \Pi t)^2) dt. \quad (4.1)$$

The apparent denominator in $\mathcal{R}$ cancels precisely when the finite jet h satisfies the cancellation equation. Take

$$\theta = 2 + \eta, \quad h(A_0) = \theta + (4\theta - 6)A_0.$$

Since $\theta^2 - 4\theta + 6 = 0$, the map

$$F^{\text{can}} = (\Pi, Q, \mathcal{R})$$

is polynomial and has determinant -1 . Its inverse equation at a target (Π, Q, R) is

$$T - \frac{Q^2}{2}T^2 + \frac{2\Pi Q}{3}T^3 - \frac{\Pi^2}{4}T^4 - R. \quad (4.2)$$

At $(\Pi, Q, R) = (4 + \eta, 3, R)$, equation (4.2) is exactly Ψ_R .

The same pencil is tangent-admissible:

$$\Psi_R(1) - \Psi_R(0) = \Psi'_R(0) = 1,$$

$$\Psi'_R(0) - \Psi'_R(1) = -1 + 2\eta,$$

$$\Psi''_R(1) - 2(\Psi'_R(1) - \Psi'_R(0)) = -5 - 8\eta.$$

Its weighted seed is fixed and boundary-clean:

$$H(T) = T^2(T - 1) \left(\frac{9}{2} - \left(\frac{7}{2} + 2\eta \right) T \right). \quad (4.3)$$

The corresponding weighted target is

$$q_R^{\text{wt}} = \left(\frac{-R}{-1 + 2\eta}, -1, 1 \right). \quad (4.4)$$

The fixed quadratic-gauge seed is

$$G(T) = \Psi_R(T) + R,$$

whose coefficients satisfy $g_1 g_3 g_4 \neq 0$. Its target is

$$q_R^{\text{quad}} = (1, 0, 2R). \quad (4.5)$$

The reconstruction derivatives in all three constructions are nonzero exactly when Ψ_R is squarefree. Thus every such specialization gives the same complete finite etale K -algebra.

As a polynomial over $K(R)$, Ψ_R is irreducible by the same degree-one-in- R argument as Lemma 3.2. Hilbert irreducibility gives the connected specializations. The stable separation in Section 5 now proves Theorem 1.2.

Proposition 4.1 (An explicit connected triple fiber). *At $R = -1$, the algebra $K[T]/(\Psi_{-1})$ is a quartic field.*

Proof. Reduce $2\Psi_{-1}$ at the degree-one prime $(17, \eta - 10)$. Since $10^2 \equiv -2 \pmod{17}$, this is a valid prime of K . After monic normalization the reduction is

$$f(T) = T^4 + 14T^3 + 2T^2 + 9T + 9 \in \mathbb{F}_{17}[T].$$

Exact modular powering gives

$$T^{17^4} - T \equiv 0 \pmod{f}$$

and

$$T^{17^2} - T \equiv 6T^3 - T^2 - 6T + 3 \pmod{f}, \quad \gcd(f, T^{17^2} - T) = 1.$$

Rabin's criterion proves that f is irreducible. Hence Ψ_{-1} is irreducible over K ; characteristic zero gives separability. $\square$

5 Stable inequivalence

We summarize the intrinsic boundary calculation used above. For a dominant quasi-finite map $F : U \rightarrow Y$, let

$$\overline{X}_F = \text{Norm}_Y K(U)$$

and retain the open immersion $U \hookrightarrow \overline{X}_F$ and its reduced boundary. This canonical finite-normalization package is functorial under polynomial source and target automorphisms. Identity stabilization replaces it by its product with affine space. Consequently the unit group of an intrinsic normalized stratum, relative Fitting ideals, and scheme-theoretic boundary-contact nilpotency are stable left-right invariants. Full proofs and exact boundary ledgers appear in [6].

For the three families used here, the boundary package intrinsically orders two target images:

$$Z_\Delta = \text{the vertex receiving a ramified boundary prime}, \quad Z_0 = \text{the other vertex}.$$

After deleting Z_0 , the normalized ramified target strata and their unit ranks are

construction	$\text{Norm}(Z_\Delta \setminus Z_0)$	$\text{rank}(\mathcal{O}^\times / K^\times)$
weighted boundary-clean	$\mathbb{A}^1 \times \mathbb{G}_m$	1
cancellation type (2, 1)	$\mathbb{G}_m^2$	2
quadratic gauge	$\mathbb{G}_m^2$	2

Affine-space factors introduce no nonconstant units. Unit rank therefore separates every weighted map used here from either reciprocal map.

For cancellation, write torus coordinates (Y, v) with $Q = (v + 1)Y$. Up to a Laurent monomial unit, the relative Fitting generator is

$$J_{\text{can}} = 2v - 1.$$

Its Laurent support has affine rank one. For a quartic quadratic gauge, after the standard shear and scaling, the corresponding generator contains the exponent points

$$(0, 0), \quad (1, 2), \quad (4, 3),$$

which have affine rank two. Torus automorphisms act on exponent lattices through affine $\text{GL}_2(\mathbb{Z})$ transformations, preserving this rank. Thus cancellation and quadratic gauge cannot become equivalent after stabilization.

There is also an independent contact check. The nilpotency indices of the scheme-theoretic intersection $Z_\Delta \cap Z_0$ are

$$\mu_{\text{wt}} = 1, \quad \mu_{\text{can}} = 6, \quad \mu_{\text{quad}} = 2. \quad (5.1)$$

Polynomial extension preserves the least nilpotency exponent. Equation (5.1) independently separates all three constructions.

6 Normalization, interpretation, and scope

If a displayed map has determinant d , translate its distinguished target to the origin and multiply one target coordinate by d^{-1} . This target automorphism changes the determinant to one, preserves the complete fiber, and leaves the left-right class unchanged. All maps in Theorems 1.1 and 1.2 may therefore be presented with determinant one and common fiber over the origin.

The point is not that two polynomials happen to have the same root set. The quotient-ring reconstruction in Propositions 2.1 and 2.2, and its cancellation analogue, identifies the entire scheme fiber with the same finite étale algebra. The surrounding maps are then separated by invariants of their canonical behavior at infinity.

The public literature currently treats arithmetic images, individual fibers, and generic degrees of Keller maps from several directions; see Gallagher’s exact degree atlas [1], the graded arithmetic analysis of Shaska [5], the ideal-theoretic fiber result of Miranda-Neto [3], and the Hilbert-irreducibility viewpoint of Lipton and Markakis [2]. The results here instead fix the finite étale algebra and ask how many stable Keller geometries contain it. No priority claim is made beyond the dated sources cited here.

Verification. The repository checker

`scripts/verify_common_arithmetic_fibers.py`

replays the tangent identities, boundary cleanliness, common inverse polynomials, the small rational quartic, the quadratic-field cancellation specialization, and the mod-17 Rabin certificate. Independent scripts replay the three family-specific stable boundary calculations.

AI assistance. Generative AI tools were used for symbolic experimentation, exact checker development, adversarial proof review, and manuscript preparation. The author reviewed the mathematical claims and assumes full responsibility for the contents.

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